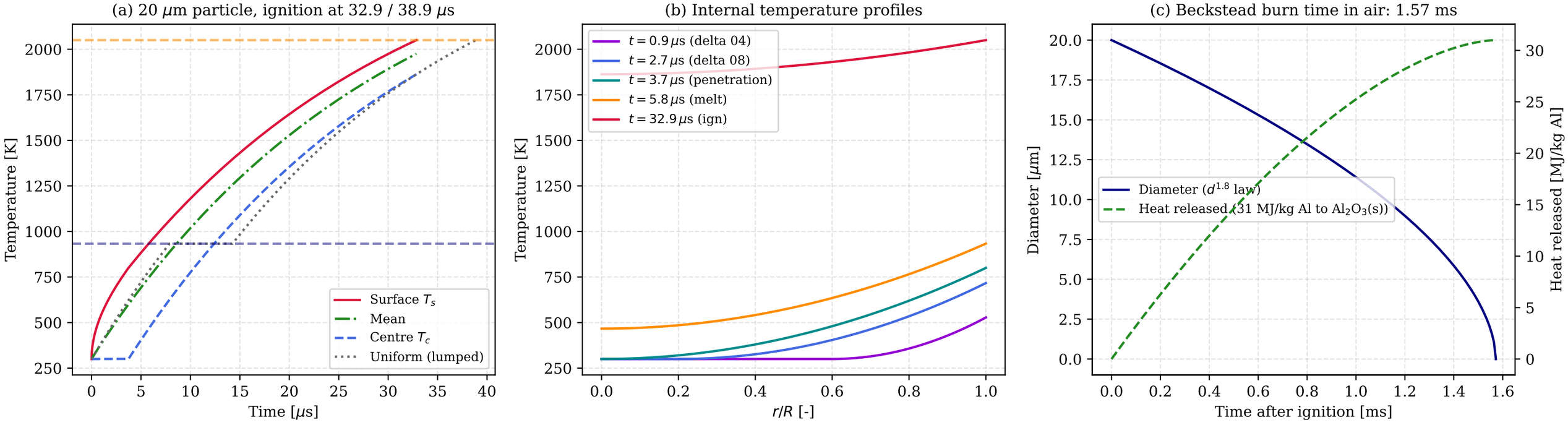
(a) 20 μm particle, ignition at 32.9 / 38.9 μs
Temperature [K]
Time [μs]
Surface T_s
Mean
Centre T_c
Uniform (lumped)
(b) Internal temperature profiles
Temperature [K]
r/R [-]
t = 0.9 μs (delta 04)
t = 2.7 μs (delta 08)
t = 3.7 μs (penetration)
t = 5.8 μs (melt)
t = 32.9 μs (ign)
(c) Beckstead burn time in air: 1.57 ms
Diameter [μm]
Heat released [MJ/kg Al]
Time after ignition [ms]
Diameter ($d^{1.8}$ law)
Heat released (31 MJ/kg Al to Al_2O_3(s))